ANNEXURE-V

Syllabi of all courses of UG programmes
B. Tech.
III Semester
[Artificial Intelligence (AI)/ Information Technology (Artificial Intelligence and Robotics)/ Artificial Intelligence (AI) and Data Science/ Artificial Intelligence (AI) and Machine Learning]
for batch admitted in 2025-26

Centre for Artificial Intelligence

PROBABILITY AND STATISTICS

(31252101/ 24252101/ 27252101/ 28252101)

COURSE OBJECTIVES

- To understand the foundational principles of probability and randomness.
- To analyze data using statistical measures and moments.
- To understand sampling and estimation techniques and perform hypothesis testing.
- To understand joint probability and random processes.

Unit-I: Classical, Empirical and Axiomatic Approaches, Laws of Compound Probability, Independence of events, Conditional Probability, Total Probability Theorem, Baye's Theorem.

Unit-II: Measures of central tendency, Standard Deviation, Skewness and Kurtosis, Expectations, Moments about origin and mean, Pearson's and Spearman's Coefficient of Correlation, Regression, Curve fitting.

Unit-III: Random Variable (Discrete & Continuous), Probability Distribution Function, Discrete Distributions: Bernoulli, Binomial & Poisson Distribution, Geometric distribution, Negative Binomial Distribution; Continuous Distributions: Normal, Exponential & Uniform Distribution.

Unit-IV: Sampling theory, Central limit theorem, Point estimation, Interval estimation, z-test, t-test, Hypothesis testing, chi-squared test, F-test, Analysis of Variance

Unit-V: Concept of Random Variables (1D & 2D), Distribution Functions: Joint, Marginal and Cumulative Distribution, Conditional Distribution Function, Concept of Random Process, Autocorrelation, stationary process, ergodic process, Markov process.

RECOMMENDED BOOKS

1. M Ray and H.S. Sharma: Mathematical Statistics, Ram Prasad Publications, 3rd Edition, 2017.
2. V.K. Kapoor, S.C. Gupta: Statistical Methods, S. Chand & Company, 11th Edition, 2018.
3. T. Veerarajan: Probability, Statistics and Random Processes, McGraw-Hill, 3rd Edition, 2008.
4. S. M. Rose: Introduction to Probability Models, Elsevier, 10th Edition, 2011.

COURSE OUTCOMES

After completion of this course, the students would be able to:

CO1	Apply probability laws and Bayes' theorem to model uncertainty.
CO2	Compute statistical measures like mean, variance, and skewness.
CO3	Determine probability distributions for AI models.
CO4	Apply sampling techniques and hypothesis testing.
CO5	Interpret joint random variables and random processes.

CO-PO Mapping Matrix														CO-WK Mapping Matrix								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	WK1	WK2	WK3	WK4	WK5	WK6	WK7	WK8	WK9
CO1	3	2									1	2	2	2	2	3	3				1	
CO2	3	3	1	1	1						1	3	3	2	2	3	3	1	1		1	
CO3	3	3		2							1	3	3	2	2	3	3				2	
CO4	3	3		3		1	1					3	3	2	2	3	3	1		1	2	1
CO5	3	3	1	2	1	1	1	1	1	1	2	3	3	2	2	3	3	1	1	1	2	1

Centre for Artificial Intelligence

DESIGN & ANALYSIS OF ALGORITHMS

(31252102/ 24252102/ 27252102/ 28252102)

COURSE OBJECTIVES

- To introduce the topic of algorithms as a precise mathematical concept.
- To study the techniques like recursion, divide and conquer, dynamic programming, greedy approach, backtracking and branch and bound.
- To practice their skills on many well-known algorithms and data structures designed to solve real-life problems.

Unit I: Introduction to Computational Model – Algorithms and its Importance, Recurrences and Asymptotic Notations, Mathematical Analysis of Non-Recursive and Recursive Algorithms.
Unit II: Divide and Conquer Method – Introduction and its Examples such as Finding the Maximum and Minimum, Binary Search, Merge Sort, Quick Sort and Strassen's Matrix Multiplication. Greedy Method: Introduction, Characteristics, Examples of Greedy Methods such as Single-Source Shortest Paths, Minimum Cost Spanning Trees: Prim's and Kruskal's Algorithm, Knapsack Problem, Dijkstra's Single Source Shortest Path Algorithm, Optimal Storage on Tapes.
Unit III: Dynamic Programming – Introduction, The Principle of Optimality, Examples of Dynamic Programming Methods such as – 0/1 Knapsack, Traveling Salesman Problem, Floyd-Warshall's All Pairs Shortest Path, Longest Common Subsequence and Reliability Design, Matrix Chain Multiplication.
Unit IV: Backtracking – Concept and its Examples like 4-Queens Problem, Knapsack problem Hamiltonian Circuit Problem, Graph Coloring Problem etc. Branch & Bound: Introduction and its Examples like - Traveling Salesperson Problem etc. NP-Completeness: Introduction, Class P and NP, Polynomial Reduction, NP-Hard and NP- Complete Problems.
Unit V: Algorithms in Machine Learning and Data Science – Gradient Descent and Stochastic Gradient Descent. Parallel and Distributed Algorithms: MapReduce. Algorithms in Blockchain: Proof-of-Work and Proof-of-Stake Algorithms.

RECOMMENDED BOOKS

1. Fundamentals of Computer Algorithms, Horowitz & Sahani, Universities press.
2. Introduction to Algorithms, Coremen Thomas, Leiserson CE, Rivest RL, PHI.
3. Design & Analysis of Computer Algorithms, Ullmann, Pearson.
4. Algorithm Design, Michael T Goodrich, Robarto Tamassia, Wiley India.

COURSE OUTCOMES

After completion of the course students will be able to:

CO1	Explain algorithm complexity using asymptotic notations .
CO2	Implement divide & conquer and greedy algorithmic approaches for problem solving.
CO3	Develop dynamic programming solutions for optimization problems.
CO4	Solve problems using backtracking and branch & bound, and classify problems under NP, NP-Complete, and NP-Hard.
CO5	Design algorithms used in machine learning, blockchain, distributed systems, and quantum computing.

CO-PO Mapping Matrix														CO-WK Mapping Matrix								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	WK1	WK2	WK3	WK4	WK5	WK6	WK7	WK8	WK9
CO1	3	3		1								2	2	2	2	3	3					
CO2	3	3	2		2							2	2	2	3	3	3	1	2			
CO3	3	3	3	2								2	2	2	2	3	3	2			1	
CO4	2	3		3				1	1	1		2	2	2	2	2	2				2	
CO5		2		2	2	1	1	1	1	1	3	3	3	1	1	1	1		2	1	2	1

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CONTROL SYSTEMS

(24252103)

COURSE OBJECTIVES

- To understand fundamental concepts of control systems and mathematical modeling of the system.
- To understand the concepts of time response and frequency response analysis of Control Systems.
- To understand the concepts of state variable models, controllability and observability as applicable to linear time invariant systems

Unit I: Introduction to Laplace Transformation, Control system modeling: Basic Elements of Control System, Open loop and Closed loop systems, Transfer function, Modelling of Electric systems, Translational and rotational mechanical systems, Block diagram reduction Techniques, Signal flow graph.
Unit II: Time response analysis: Standard test signals, time response of first order systems, Impulse and Step Response analysis of second order systems, time domain specifications, steady state response, steady state errors and error constants, effects of P, PI, PD and PID.
Unit III: Stability analysis: stability, Routh-Hurwitz Criterion, Root Locus Technique, Construction of Root Locus, effects of adding poles and zeros to G(s)H(s) on the root loci.
Unit IV: State variable analysis: State space representation of Continuous Time systems, State equations, Transfer function from State variable representation, Solutions of the state equations, canonical variable diagonalization, system analysis by transfer function and state space methods for continuous systems. Concept of controllability and observability, design of state feedback controllers, Pole placement by state feedback, set point tracking controller, Compensators – Lead, Lag, and Lead Compensators.
Unit V: Intelligent Control Systems, Model Predictive Control (MPC), Adaptive and Self-Tuning Controllers, Control in Autonomous Systems, Cyber-Physical Systems (CPS) and Networked Control Systems (NCS), Data-Driven and Machine Learning-Based Control.

RECOMMENDED BOOKS

1. Automatic control systems, Benjamin. C. Kuo, Prentice Hall of India.
2. Modern Control Engineering, Kotsuhiko Ogata, Prentice Hall of India.
3. Control Systems Engineering, I.J. Nagrath & M. Gopal, New Age Pub. Company.
4. Control System – Principles and Design, M. Gopal, Tata McGraw Hill.
5. Feedback and Control Systems, Schaum's Outline Series, Tata McGraw Hill.
6. Digital Control and State Variable Methods, M. Gopal, Tata Mcgraw Hill.

COURSE OUTCOMES

After completion of the course students will be able to:

CO1: Explain Laplace transforms, control system modeling, and block diagram simplification.
CO2: Analyze time response of systems and evaluate P, PI, PD, and PID controller effects.
CO3: Examine system stability using Routh-Hurwitz and construct root locus plots.
CO4 Develop state-space models and design controllers using controllability and observability.
CO5: Explore intelligent, adaptive, and machine learning-based control techniques.

CO-PO Mapping Matrix														CO-WK Mapping Matrix								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	WK1	WK2	WK3	WK4	WK5	WK6	WK7	WK8	WK9
CO1	3	2	1									3	2	2	2	2	2					
CO2	3	3		2								3	2	2	2	3	3				1	
CO3	3	3		3								3	2	2	2	3	3				2	
CO4	3	3	3	2	2			1	1	1		3	3	2	3	3	3	2	2		1	
CO5		2	2		2	1	1	1	1	1	3	3	3	1	1	1	1	2	2	1	2	1

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DATABASE MANAGEMENT SYSTEM

(31252103/ 27252103/ 28252103)

COURSE OBJECTIVES

- To understand the different issues involved in the design and implementation of a database system.
- To study the physical and logical database designs, database modelling, relational, hierarchical and network models.
- To understand and use data manipulation language to query, update and manage a database.

Unit I: DBMS – Database Approach v/s Traditional File Approach, Advantages of Database System, Database Users and Administrator, Database System Environment, Application Architectures, Schemas, Instances, Data Independence, Data Models: Hierarchical Data Model, Network Data Model & Relational Data Model, Comparison between Models. Entities and Relationship Model: Entity types, Entity sets, Attributes and Keys, Relationship Types and Sets, Constraints, Design issue, E-R Diagram, Weak Entity Sets.

Unit II: Relational Model – Structure of Relational Databases: Relation, Attribute, Domain, Tuples, Degree, Cardinality, Views, Database Relations, Properties of Relations, Attributes, Keys, Attributes of Relation, Domain Constraints, Integrity Constraints. Relational Algebra: Concepts and Operations: Select, Project, Division, Intersection, Union, Division, Rename, Join etc.

Unit III: SQL – Purpose of SQL, Data Definition Language (DDL) Statements, Data Manipulation Language (DML) Statements Update Statements & Views in SQL, Data Control Language (DCL), Triggers. Relational Database Design: Purpose of Normalization, Data Redundancy and Update Anomalies, Functional Dependency, Process of Normalization, Various Normal Forms: 1NF, 2NF, 3NF, BCNF, Decomposition, Desirable Properties of Decomposition: Dependency Preservation, Lossless Join, Problems with Null Valued & Dangling Tuple, Multivalued Dependencies.

Unit IV: Transaction Management – Transaction Concept, Transaction State, Concurrent Executions, Serializability: Conflict and View Serializability, Concurrency Control: Lock-Based Protocol, Recovery: Log-Based Recovery.

Unit V: Big Data and Distributed Databases: CAP theorem, eventual consistency, sharding. NoSQL and NewSQL Databases: Document (MongoDB), Key-Value (Redis), Column-family (Cassandra), Graph (Neo4j), Google Spanner, CockroachDB. Real-Time Data Processing and Streaming Databases: Apache Kafka, Apache Flink.

RECOMMENDED BOOKS

1. Database System Concepts, Abraham Silberschatz Henry F. Korth S. Sudarshan, McGraw-Hill 6th Edition.
2. Database Management System, Raghu Ramakrishnan Johannes Gehrke, McGraw Hill 3rd Edition.
3. Fundamentals of Database System, Elmasri & Navathe, Addison-Wesley Publishing, 5th Edition.
4. An Introduction to Database Systems, Date C. J, Addison-Wesley Publishing, 8th Edition.

COURSE OUTCOMES

After completion of the course students will be able to:

CO1 Differentiate between traditional file systems and the database approach.
CO2 Construct relational schemas and perform relational algebra operations.
CO3 Develop SQL queries for data definition, manipulation, and control, and analyze relational schemas for normalization.
CO4 Analyse transaction processing, concurrency control and recovery mechanisms in database systems.
CO5 Compare traditional RDBMS with modern databases.

CO-PO Mapping Matrix														CO-WK Mapping Matrix								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	WK1	WK2	WK3	WK4	WK5	WK6	WK7	WK8	WK9
CO1	3	2										1	1	2	2	2	2					
CO2	3	3	2		1							1	2	2	2	3	3	1	1			
CO3	2	3	3		2							2	2	2	2	2	2	2	2			
CO4	2	3		3				1	1	1		2	2	2	2	2	2				2	
CO5		2			2	1	1	1	1	1	3	2	3	1	1	1	1		2	1	2	1

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COMPUTER NETWORKS

(31252104/ 24252104/ 27252104/ 28252104)

COURSE OBJECTIVES

- Familiarize the student with the basic taxonomy and terminology of the computer networking.
- Provide detailed knowledge about various layers, protocols and devices that facilitate networking.
- Enable Students to deal with various networking problems such as flow control, error control and congestion control.

Unit I: Network Standardization- OSI Reference Model & TCP/IP Reference Model. Performance Criteria- Bandwidth, Throughput, Propagation Time & Transmission Time. Physical Layer: Network Topologies- Bus, Ring, Star & Mesh, Switching- Circuit Switching, Message Switching & Packet Switching, Multiplexing: FDM – Frequency Division Multiplexing, WDM – Wavelength Division Multiplexing & TDM – Time Division Multiplexing.

Unit II: Data Link Layer – Introduction, Design Issues, Services, Framing, Error Control, Flow Control, ARQ Strategies, Error Detection and Correction, Parity Bits, Cyclic Redundancy Code (CRC), Hamming Codes, MAC Sub Layer- The Channel Allocation Problem, Pure ALOHA, Slotted ALOHA, CSMA, CSMA/CD, IEEE 802.3, IEEE 802.4 and IEEE 802.5.

Unit III: Network Layer & Transport Layer – Introduction, Design Issues, Services, Routing- Distance Vector Routing, Hierarchical Routing & Link State Routing, Shortest Path Algorithm- Dijkstra's Algorithm & Floyd–Warshall's Algorithm, Flooding, Congestion Control- Open Loop & Closed Loop Congestion Control, Leaky Bucket & Token Bucket Algorithm. Connection Oriented & Connectionless Service, IP Addressing.

Unit IV: Presentation, Session & Application Layer – Introduction, Design Issues, Presentation Layer- Translation, Encryption- Substitutions and Transposition Ciphers, Compression- Lossy and Lossless. Session Layer – Dialog Control, Synchronization. Application Layer- Remote Login, File Transfer & Electronic Mail.

Unit V: Software-Defined Networking (SDN), Edge and Fog Computing in Networking, Quantum Networking (Introductory Concepts), Zero Trust Architecture (ZTA), Next-Generation Internet Protocols: Segment Routing, Multipath TCP (MPTCP), QUIC.

RECOMMENDED BOOKS

1. Data Communication and Networking, Behrouz A. Forouzan, McGraw Hill.
2. Computer Networks, Andrew S. Tanenbaum, Pearson Education India.
3. Computer Networks and Internets, Douglas E. Comer, Pearson India.

COURSE OUTCOMES

After completion of the course students will be able to:

CO1 Illustrate network topologies, switching and multiplexing techniques.

CO2 Analyze data link layer protocols for error control and channel access.

CO3 Evaluate routing algorithms and congestion control mechanisms.

CO4 Analyze the functions and services of presentation, session, and application layers.

CO5 Evaluate emerging networking technologies and next-generation internet protocols.

CO-PO Mapping Matrix														CO-WK Mapping Matrix								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	WK1	WK2	WK3	WK4	WK5	WK6	WK7	WK8	WK9
CO1	3	2										1	2	2	2	2	2					
CO2	3	3		2								1	2	2	2	3	3				1	
CO3	2	3	2	2	2							1	2	2	2	2	2	1	2		1	
CO4	3	2			1			1	1	1		1	2	2	2	2	2		1			
CO5		2			2	1	1	1	1	1	3	2	3	1	1	1	1		2	1	2	1

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OPERATING SYSTEMS

(31252105/ 24252105/ 27252105/ 28252105)

COURSE OBJECTIVES

- To provide basic knowledge of computer operating system structures and functioning.
- To compare several different approaches to memory management, file management and process management.
- To understand various problems related to concurrent operations and their solutions.

Unit I: Basics of Operating System: Generations, Types, Structure, Services, System Calls, System Boot, System Programs, Protection and Security. Process Management: Process Concepts, Process States, Process Control Block, Scheduling-Criteria, Scheduling Algorithms and their Evaluation, Threads, Threading Issues.

Unit II: Process Synchronization: Background, Critical-Section Problem, Peterson's Solution, Synchronization Hardware, Semaphores, Classic Problems of Synchronization, Monitors. Deadlock: System Model, Deadlock Characterization, Deadlock Prevention, Detection and Avoidance, Recovery from Deadlock.

Unit III: Memory Management: Main Memory, Swapping, Contiguous Memory Allocation, Paging, Structure of Page Table, Segmentation, Virtual Memory, Demand Paging, Page Replacement Algorithms, Allocation of Frames, Thrashing.

Unit IV: Storage Management: Mass-Storage Structure, Overview, Disk Structure, Disk Attachment, Disk Scheduling. File System Interface: The Concept of a File, Access Methods, Directory Structure, File System Structure, Allocation Methods, Free-Space Management.

Unit V: Containerization and Operating System-Level Virtualization, Kubernetes and Container Orchestration, Secure Boot, Trusted Execution Environments, OS-level sandboxing and isolation, Microkernel and Exokernel Architectures, GPU/TPU scheduling and driver integration, Lightweight OS (KaiOS, Tizen, PostmarketOS).

RECOMMENDED BOOKS

1. Operating System Concepts, Silberschatz, Ninth Edition, Wiley Publication.
2. Operating Systems, internals and Design Principles, Stallings, Seventh Edition, Pearson Publication.
3. Modern Operating Systems, Tanenbaum, Fourth Edition. Pearson Publication.

COURSE OUTCOMES

After completion of the course students will be able to:

CO1	Describe the fundamental concepts and services of operating systems
CO2	Demonstrate solutions to process synchronization problems and analyze different approaches to deadlock detection, avoidance, and recovery.
CO3	Implement memory management techniques, and evaluate page replacement algorithms.
CO4	Analyze different file systems, storage structures, allocation mechanisms and free-space management methods.
CO5	Evaluate the impact of modern OS-level technologies on performance and security.

CO-PO Mapping Matrix														CO-WK Mapping Matrix								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	WK1	WK2	WK3	WK4	WK5	WK6	WK7	WK8	WK9
CO1	3	2										1	2	2	2	2	2					
CO2	3	3		2								1	2	2	2	3	3				1	
CO3	2	3	1	3	1							1	2	2	2	2	2		1		2	
CO4	2	3		3				1	1	1		1	2	2	2	2	2				2	
CO5		2			2	2	1	1	1	1	3	2	3	1	1	1	1	1	2	2	2	1

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CYBER SECURITY
(31252111/ 24252111/ 27252111/ 28252111)

COURSE OBJECTIVES

- To introduce the basic concepts of cybersecurity.
- To make students aware of various types of cyber threats, vulnerabilities, security policies and cybersecurity tools.
- To build basic skills for protecting information systems.

Unit I: Introduction to Cyber Security: Overview of Cyber Security, Goals of Cyber Security (Confidentiality, Integrity, Availability), Types of cyber attacks: Phishing, Malware, Ransomware, Social Engineering, Malicious Softwares. Hacker and its types. Real-world incidents and their impact, Cyber Ethics and Legal Aspects.

Unit II: Basics of Networking: Internetworking devices, Topologies OSI and TCP/IP models, IP address, DNS, TCP, IP, HTTP, HTTPS, Web Browser, Web Server.

Unit III: Security Mechanisms: Firewalls, Anti-virus, Intrusion Detection Systems (IDS), Intrusion Prevention Systems (IPS), Encryption and Decryption: Symmetric and Asymmetric, Cryptanalysis, Digital Signature, Authentication: Passwords, Biometrics, Multi-Factor Authentication.

Unit IV: System and Application Security: Operating System security basics. Securing mobile devices and apps. Web application vulnerabilities: SQL Injection, XSS, CSRF. Secure coding practices. Cybercrime, Forensics, and Incident Response: Types of cybercrimes: Identity Theft, Financial Fraud, Cyberbullying. Basics of digital forensics. Cyber law and IT Act (India) overview. Incident response lifecycle and reporting.

Unit V: Cyber Hygiene and Best Practices: Cyber hygiene: Safe browsing, regular updates, backups. Strong password creation and management. Social media safety. Roles of individuals and organizations in ensuring security.

RECOMMENDED BOOKS

1. "Cybersecurity for Beginners" by Raef Meeuwisse – Wiley
2. "Cyber Security: Understanding Cyber Crimes, Computer Forensics and Legal Perspectives" by Nina Godbole and Sunit Belapure – Wiley India
3. "Computer Security: Principles and Practice" by William Stallings and Lawrie Brown – Pearson
4. "Introduction to Cyber Security" by Chwan-Hwa (John) Wu and J. David Irwin – CRC Press
5. "Cybersecurity Essentials" by Charles J. Brooks, Christopher Grow, Philip Craig, Donald Short – Wiley

COURSE OUTCOMES

After completion of this course, the students would be able to:

CO1	Describe fundamental concepts of cyber security and identify common cyber threats and legal implications.
CO2	Classify basic networking concepts.
CO3	Demonstrate common security mechanisms used to protect digital data.
CO4	Analyze cybercrime scenarios and vulnerabilities in systems, and outline procedures for incident response and digital forensics.
CO5	Formulate cyber hygiene strategies and practice safe online behavior to minimize cyber risks.

CO-PO Mapping Matrix														CO-WK Mapping Matrix								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	WK1	WK2	WK3	WK4	WK5	WK6	WK7	WK8	WK9
CO1	3	2					2					1	1	2	2	2	2					2
CO2	3	2										1	2	2	2	2	2					
CO3	2	2	2		3		2					1	2	1	2	2	2	1	3			2
CO4		3		3		2	2	1	1	1		2	2	2	1	2	2	1		2	2	2
CO5			1			2	3	1	1	1	2	1	1	1				2		2	1	3

ANNEXURE-VI

Experiment list/ Lab manual for all the Laboratory Courses and Macro Project-I to be offered in B. Tech. III Semester [Artificial Intelligence (AI)/ Information Technology (Artificial Intelligence and Robotics)/ Artificial Intelligence (AI) and Data Science/ Artificial Intelligence (AI) and Machine Learning] under the flexible curriculum

(Batch admitted in academic session 2025 – 26)

Centre for Artificial Intelligence

PROBLEM SOLVING THROUGH PYTHON PROGRAMMING
(31252106/ 24252106/ 27252106/ 28252106)

List of Programs

1. Core Logic & Advanced Data Types
   - Prime Sieve: Generates prime numbers efficiently up to N for cryptographic foundations.
   - List Comprehension Matrix Transpose: Converts rows to columns using nested comprehensions instead of loops.
   - Dictionary-Based Sparse Matrix: Represents memory-efficient matrices using coordinate tuples as dictionary keys.
   - Anagram Grouping: Groups lists of words into anagram categories using sorted strings as hash map keys.
   - Two-Pointer Subarray Search: Finds a continuous subarray summing to a target value in N time.
2. Object-Oriented Programming (OOP)
   - Neural Network Layer Class: Implements encapsulation with weights, biases, and a forward propagation method.
   - Abstract Model Evaluator: Uses inheritance to build a base evaluation class extended by specific metric classes.
   - Polymorphic Activation Functions: Creates different activation classes (ReLU, Sigmoid) sharing a common execution interface.
   - Property Decorator Encapsulation: Validates learning rates and hyperparameter values using getter and setter properties.
   - Operator Overloading Vector: Overloads __add__ and __mul__ to perform element-wise coordinate operations.
3. Robust Error & Exception Handling
   - Custom Dataset Exception: Raises custom errors when input files contain missing or corrupted data shapes.
   - Safe Division Evaluator: Handles ZeroDivisionError and TypeError dynamically during mathematical metric updates.
   - Resource Monitor Context Manager: Uses with statements to guarantee system memory cleanup during training loops.
   - Retry Mechanism Decorator: Catches network timeouts and retries API data fetches up to three times.
4. Advanced File & Data Pipeline Handling
   - Batch Image Path Generator: Parses nested directories using pathlib to filter images for model queues.
   - Streaming Large Log Files: Uses generators to read multi-gigabyte log files line-by-line without memory crashes.
   - JSON Configuration Merger: Validates, updates, and writes nested hyperparameters into persistent JSON storage.
   - Binary Model Serialization: Uses pickle or marshal to save and reload trained state configurations.
5. Numerical Computing with NumPy
   - Dataset Normalization: Converts raw matrices into Z-score distributions using vectorized mean and standard deviation.
   - One-Hot Encoding Vectorization: Transforms categorical integer labels into binary vectors without loop structures.
   - Boolean Array Masking: Extracts, filters, and replaces matrix outliers based on standard deviation thresholds.
   - Broadcasting Image Adjustments: Multiplies a 3D image tensor by a 1D color-channel scale vector.
   - Linear System Solver: Uses numpy.linalg to solve simultaneous equations representing linear regressions.

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COURSE OUTCOMES

After completion of the course students will be able to:

CO1	Implement structured Python programs utilizing advanced data structures, control flows, and modular programming constructs to solve algorithmic problems.
CO2	Model real-world software entities using Object-Oriented Programming (OOP) concepts to build scalable applications.
CO3	Analyze robust file handling pipelines and custom exception frameworks to handle runtime errors and manage unstructured data streaming safely.
CO4	Design vectorized data preprocessing algorithms using NumPy to manipulate high-dimensional matrices and mathematical datasets for artificial intelligence workflows.

CO-PO Mapping Matrix														CO-WK Mapping Matrix								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	WK1	WK2	WK3	WK4	WK5	WK6	WK7	WK8	WK9
CO1	3	3	3	2	3						2	2	2	2	3	3	3	2	3		2	
CO2	2	3	3	2	2							2	3	2	2	2	2	2	2		1	
CO3	3	3	3	3	3						3	3	3	2	3	3	3	2	3		3	
CO4	2	3		2	2			1	1	1		2	2	2	2	2	2		2		1	
CO5		2	3		3	1	1	1	1	1	2	2	3	1	2	1	1	2	3	1	1	1

Centre for Artificial Intelligence

DESIGN & ANALYSIS OF ALGORITHMS LAB
(31252107/ 24252107/ 27252107/ 28252107)

List of Programs

1. Program to calculate time complexity of different sorting algorithms
2. Implement and compare iterative vs recursive Fibonacci computation
3. Measure execution time of basic operations (e.g., searching in arrays)
4. Implement Merge Sort and Quick Sort
5. Implement Binary Search (recursive and iterative)
6. Find Maximum and Minimum in an array using divide and conquer
7. Design and implement solution of following problems using Greedy Algorithms:
    a. Fractional Knapsack Problem
    b. Activity Selection Problem
    c. Huffman Coding
    d. Job Sequencing with Deadlines
    e. Minimum Spanning Tree: Kruskal's and Prim's algorithms
8. Design and implement solution of following problems using Dynamic Programming Approach:
    a. 0/1 Knapsack Problem
    b. Longest Common Subsequence (LCS)
    c. Matrix Chain Multiplication
9. Traverse a graph using Depth-First Search (DFS) and Breadth-First Search (BFS) algorithms.
10. Find a single source shortest path, in a weighted graph, using Dijkstra's Algorithm.
11. Find all-pair shortest paths, in a weighted graph, using Floyd-Warshall Algorithm.
12. Implement the classic word count problem using the MapReduce framework.
13. Implement gradient descent to minimize MSE for a linear regression problem.

COURSE OUTCOMES

After completion of the course students will be able to:

CO1	Apply algorithmic design techniques to solve computational problems.
CO2	Analyze real-world problems using different algorithmic design approaches.
CO3	Evaluate algorithm efficiency and understand computational intractability.

CO-PO Mapping Matrix														CO-WK Mapping Matrix								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	WK1	WK2	WK3	WK4	WK5	WK6	WK7	WK8	WK9
CO1	3	3	2		2							2	2	2	3	3	3	1	2			
CO2	3	3	2	2	2							2	2	2	3	3	3	1	2		1	
CO3	3	3	3	3	2							2	2	2	3	3	3	2	2		2	
CO4		2	2		2			1	1	1	2	2	2	1	1	1	1	1	2		1	
CO5		2	3		3	1	1	1	1	1	3	2	2	1	2	1	1	2	3	1	2	1

Centre for Artificial Intelligence

DATABASE MANAGEMENT SYSTEM LAB

1. SQL Queries: SELECT queries using WHERE, BETWEEN, LIKE, ORDER BY, GROUP BY, HAVING.
2. DDL Commands: CREATE, ALTER, DROP tables. Defining constraints (PK, FK, UNIQUE, CHECK, NOT NULL).
3. DML Commands: INSERT, UPDATE, DELETE operations using sample datasets.
4. SQL Joins: INNER JOIN, LEFT JOIN, RIGHT JOIN, FULL JOIN queries.
5. Set Operations: UNION, INTERSECT, MINUS/EXCEPT operations.
6. Aggregate Functions: COUNT, SUM, AVG, MIN, MAX with grouping.
7. Subqueries & Nested Queries: Single-row, multi-row, and correlated subqueries.
8. Views: Creating, updating, and deleting views.
9. Indexes & Sequences: Creating and using indexes for optimization. Creating sequences (if supported).
10. PL/SQL Basics: Anonymous blocks demonstrating variables, loops, and conditional statements.
11. Stored Functions: Writing user-defined functions (UDFs).
12. Stored Procedures: Creating and invoking stored procedures.
13. Cursors: Explicit cursor to iterate over table rows.
14. Exceptions in PL/SQL: Predefined and user-defined exceptions.
15. Database Triggers: Row-level & statement-level triggers (BEFORE/AFTER INSERT/UPDATE/DELETE).
16. Transactions & Concurrency: Demonstration of COMMIT, ROLLBACK, SAVEPOINT. Concurrent update simulation (using two sessions).

COURSE OUTCOMES

After completion of this course, the students would be able to:

CO1	Apply basic SQL commands (DDL, DML, DCL, TCL) to create and manage relational database objects.
CO2	Execute complex SQL queries involving joins, subqueries, set operations, indexing, and views.
CO3	Develop PL/SQL blocks including functions, procedures, cursors, exceptions, and triggers to implement business logic.
CO4	Evaluate transaction management, concurrency control, and integrity constraints in relational databases.

CO-PO Mapping Matrix														CO-WK Mapping Matrix								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	WK1	WK2	WK3	WK4	WK5	WK6	WK7	WK8	WK9
CO1	2	2	3		3							1	2	1	2	2	2	2	3			
CO2	2	3	3		2							1	2	2	2	2	2	2	2			
CO3		2	2	3	2							1	2	1	1	1	1	1	2		2	
CO4		3	3		2			1	1	1		1	2	1	2	2	2	2	2			
CO5		2			2	1	1	1	1	1	2	2	2	1	1	1	1		2	1	1	1

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Design Thinking Lab

1. Introduction to CAD Modeling for Robotics Applications
2. 2D Sketching and Parametric Design using CAD Software
3. Design of Robot Chassis using CAD Tools
4. 3D Modeling of Robotic Arm Links and Joints
5. CAD-based Design and Assembly of Mobile Robot
6. Design of Robotic Gripper Mechanism
7. STL File Generation and Slicing for 3D Printing
8. Fabrication of Robot Components using 3D Printer
9. Assembly and Testing of 3D Printed Robot Prototype
10. Project: Design Thinking and Rapid Prototyping of a Functional Robot System

COURSE OUTCOMES

After completion of this course, the students would be able to:

CO1	Explain the fundamentals of CAD modeling and 3D printing for robotic prototyping applications.
CO2	Design robotic components and assemblies using parametric CAD tools.
CO3	Develop and optimize robot structures considering functionality, manufacturability, and mechanical constraints.
CO4	Evaluate STL models, perform slicing, and fabricate robotic parts using 3D printing technology.
CO5	Develop design-thinking approaches for industry-oriented robotic systems.

CO-PO Mapping Matrix														CO-WK Mapping Matrix								
	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PSO1**	**PSO2**	**WK1**	**WK2**	**WK3**	**WK4**	**WK5**	**WK6**	**WK7**	**WK8**	**WK9**
CO1	2	2	3	2	3							2	3	1	2	2	2	2	3		1	
CO2		2	3	2	3	1		2				3	3	1	2	1	1	2	3	1	1	
CO3			3	2	3			2	2			2	2		1			2	3		1	
CO4			3	3	3					2		2	2		1			2	3		2	
CO5			2		2	1	1	2	1	2	2	1	1		1			2	2	1	1	1

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MACRO PROJECT-I

(31252109/ 24252109/ 27252109/ 28252109)

List of Programs

1. Design an ER-diagram for the following:
    a. Library Management System
    b. Online Shopping Database
    c. Examination Result Processing System
2. Design a Student Marksheet System
3. Implement Contact Book using Dictionary
4. Design and implement To-Do List CLI Tool
5. Create a simple Quiz Game
6. Develop a basic ATM Simulator.
7. Develop a Personal Finance Tracker to track income/expenses using Python.
8. Design a Text Analyzer (counts frequency, finds longest word, etc.).
9. Design a Real-Time Chat Application (Console/GUI) using Python sockets or Flask for communication.
10. Create a system which implements route optimizer using Dijkstra algorithm.
11. Create an exam scheduling system using Graph Coloring
12. Develop a File Compressor using Huffman Coding
13. Create an Auto-Correct or Spell Checker using Edit Distance
14. Implement Backup and recovery handling in DBMS.
15. Design a Smart Parking System using algorithmic slot allocation (first-fit, best-fit), user booking via Python interface, and DB to track occupancy.
16. Create an Auction System with Bid Optimization to track users and bids in DB; applies greedy algorithms for winner selection based on utility or price/time factors.
17. Develop a Real-Time Voting App with Conflict Resolution using python and uses Paxos/Raft-inspired algorithm to ensure vote consistency across distributed systems (can simulate using threads).

Note: use SQL at the backend for storing data.

COURSE OUTCOMES

After completion of the course students will be able to:

CO1	Develop database-driven applications using SQL.
CO2	Implement algorithmic solutions using Python for computational and optimization problems.
CO3	Integrate database and algorithmic logic in a cohesive application to solve real-world problems.

CO-PO Mapping Matrix														CO-WK Mapping Matrix								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	WK1	WK2	WK3	WK4	WK5	WK6	WK7	WK8	WK9
CO1	2	2	3	2	3							2	3	1	2	2	2	2	3		1	
CO2		2	3	2	3	1		2		1		3	3	1	2	1	1	2	3	1	1	
CO3			1			1	1	3	2	2	1	1	1					1		1		1